

AGENTIC AI LAB INTERNAL – 1 (21-092026)

1Q. Text to SQL workflow , Build an end to end LLM with retrieval and query generation Output:

The image displays two screenshots of a web application titled "AI Text-to-SQL Workflow". The interface is dark-themed and includes a sidebar on the left with a "Text-to-SQL" header and a "Workflow" section. The workflow steps are: 1. User Question (Natural language input), 2. Retrieval (Relevant database schema is retrieved), 3. SQL Generation (The question is converted into SQL), and 4. Result (Display database answer).

The main content area shows the "AI Text-to-SQL Workflow" title and a description: "Convert natural language questions into SQL queries using Retrieval + LLM-style Query Generation + Execution." Below this, there are four numbered steps: 1. Retrieval (Retrieve relevant database schema), 2. SQL Generation (Convert natural language to SQL), 3. Execution (Execute generated SQL), and 4. Result (Display database answer).

The "Ask Your Database" section prompts the user to "Enter your question in natural language:" with an example: "Example: Which product generated the highest revenue?".

The second screenshot shows the "Ask Your Database" section with a text input field containing "what is highest revenue?". Below the input field, there are several example queries: "What is the total revenue?", "Which product generated ...", "How many customers are ...", "How many orders are there?", "List all products", "Show customers", "Show revenue by product", and "Show orders per customer". A "Run Text-to-SQL Workflow" button is visible, followed by a green notification bar stating "Workflow completed successfully!". Below this, the "Step 1 — Retrieved Schema" section shows a table named "orders" with a column "order_id".

Text-to-SQL

Workflow

User Question

Natural language input.

↓

Retrieval

Relevant database schema is retrieved.

↓

SQL Generation

The question is converted into SQL.

↓

Step 1 — Retrieved Schema

orders

order_id

customer_id

product_id

quantity

total_amount

order_date

Text-to-SQL

Workflow

User Question

Natural language input.

↓

Retrieval

Relevant database schema is retrieved.

↓

SQL Generation

The question is converted into SQL.

↓

Step 2 — Generated SQL

```
SELECT
  p.product_name,
  SUM(o.total_amount) AS revenue
FROM orders o
JOIN products p
  ON o.product_id = p.product_id
GROUP BY p.product_id, p.product_name
ORDER BY revenue DESC
LIMIT 1;
```

Step 3 — Query Execution

SQL query executed successfully.

	product_name	revenue
0	Laptop	225000

Step 4 — Final Result

Agentic AI Exam Help x AI Text-to-SQL x + Ask Gemini

localhost:8501

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Text-to-SQL

Workflow

- User Question
Natural language input.
- Retrieval
Relevant database schema is retrieved.
- SQL Generation
The question is converted into SQL.

```
GROUP BY p.product_id, p.product_name
ORDER BY revenue DESC
LIMIT 1;
```

Step 3 — Query Execution

SQL query executed successfully.

	product_name	revenue
0	Laptop	225000

Step 4 — Final Result

Rows Returned
1

The natural-language question was converted into SQL and executed against the database.

2d. RAG based question answering system , implement indexing , retrieval and response generation Output:

Agentic AI Exam Help x RAG Question Answering x + Ask Gemini

localhost:8501

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RAG System

Pipeline

- Document Loading
- Indexing
- Retrieval
- Response Generation

RAG-Based Question Answering System

Retrieval-Augmented Generation using Indexing • Retrieval • Response Generation

RAG Pipeline

- Documents**
Load knowledge from documents.
- Indexing**
Convert document chunks into searchable vectors.
- Retrieval**
Find relevant knowledge for the question.
- Response**
Generate an answer using retrieved context.

Knowledge base indexed successfully • 24 document chunks

Ask a Question

Pipeline

Document Loading

↓

Indexing

↓

Retrieval

↓

Response Generation

Ask a Question

Enter your question:

what is rag?

Try an example:

What is Artificial Intelligence?

What is RAG?

What are embeddings?

How does a RAG pipeline work?

What is information retrieval?

What are the advantages of RAG?

Retrieve & Generate Answer

Step 1 — Retrieved Context

> Retrieved Document 1 • Similarity: 1.00

> Retrieved Document 2 • Similarity: 0.49

> Retrieved Document 3 • Similarity: 0.49

Pipeline

Document Loading

↓

Indexing

↓

Retrieval

↓

Response Generation

Retrieved Document 2 • Similarity: 0.49

Retrieved Document 3 • Similarity: 0.49

Step 2 — Generated Response

Answer

RAG

Workflow Execution

Indexed Chunks	Retrieved Chunks	Top Similarity
24	3	1.00